HUSSAIN

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Portfolio: https://portfolio-hussain-five.vercel.app/

GitHub: github.com/Hussain-Innovator

SUMMARY

BS Software Engineering graduate specializing in Artificial Intelligence, Computer Vision, and Deep Learning. Experienced in developing and deploying AI models for image classification, medical imaging, and web-based applications using Python, PyTorch, TensorFlow, and Streamlit. Alongside AI development, I have over two years of frontend development experience and hands-on experience in data analysis and IT support. Further details, projects, and practical work can be viewed through my portfolio.

EDUCATION

BS Software Engineering

Iqra University — Karachi, Pakistan | Sep 2021 – Sep 2025

Major: Artificial Intelligence & Computer Vision
Final Year Project:
Fine-Grained Classification of Ultrasound Fetal Planes

WORK EXPERIENCE

IT Assistant

Noor Recruiting Agency — Islamabad, Pakistan | Sep 2025 – Jan 2026

- Manage and maintain IT systems, networks, and hardware/software.
- Provide technical support and troubleshoot system issues.
- Implement data security measures, backups, and system updates.
- Communicate with vendors and clients to resolve IT-related issues.

Data Analyst

Knots Solutions Pvt Ltd — Karachi, Pakistan | Jun 2025 – Sep 2025

- Collected and processed datasets for reporting and decision-making.
- Performed data analysis to identify patterns and trends.
- Assisted in creating dashboards and visual reports.
- Improved data accuracy and supported operational processes.

Freelance Frontend Developer

Remote | Jan 2023 – Aug 2024

- Developed responsive web interfaces using modern frontend tools.
- Collaborated with teams on real-world development projects.
- Communicated with clients to deliver required features.

AI Researcher & Model Developer

Freelance | Oct 2024 – Present

- Developed machine learning and computer vision models.
- Conducted experiments and research to improve model accuracy.
- Worked on medical imaging analysis projects.
- Prepared technical reports and visualizations.

TECHNICAL SKILLS

Programming
Python, JavaScript, Basic C, Java

AI & Deep Learning
PyTorch, TensorFlow, Keras, Scikit-learn, Captum (XAI), NumPy, Pandas, Matplotlib

Frontend
HTML, CSS, JavaScript, React, Responsive Web Design

Tools
Git, GitHub, Google Colab, Jupyter Notebook, VS Code, Streamlit, Hugging Face Spaces, ONNX, Runtime, Power BI, MS Office

PROJECTS

Fine-Grained Classification of Fetal Ultrasound Planes (Final Year Project)

- Built CNN model with MobileNetV2 and SE Blocks using PyTorch.
- Achieved 93.34% validation accuracy on FETAL_PLANES_DB dataset.
- Applied Explainable AI techniques including Grad-CAM and Guided Backpropagation.
- Deployed model using Streamlit application for real-time predictions.

Hybrid CNN — CIFAR-10 Image Classifier

- Designed a custom hybrid CNN architecture combining Residual Connections, Squeeze-and-Excitation (SE) Attention, and Depthwise Separable Convolutions.
- Trained the model on the CIFAR-10 dataset consisting of 60,000 labeled images across 10 object categories.
- Achieved 90.93% test accuracy, 98.71% Top-3 accuracy, and a Macro F1-score of 0.91.
- Improved model performance using MixUp augmentation, label smoothing, cosine annealing learning rate scheduling, and linear warmup.
- Converted the trained TensorFlow model to ONNX format for efficient CPU inference.
- Deployed interactive applications using Streamlit and Hugging Face Spaces with ONNX Runtime.

RESEARCH & REPORTS

- Ablation Study: Contribution of SE Block, Convolutional Layers & Pooling Strategy in FetalNet
- Hybrid CNN Architecture for CIFAR-10 Image Classification — Technical Report

CERTIFICATIONS

- Python for Data Science, AI & Development
- MS Office
- InPage
- Research Paper Writing
- Jira Fundamentals
- English for Workplace Communication

ADDITIONAL INFORMATION

- Activities: Cricket, Swimming,Volleyball
- Soft Skills: Time Management, Independent Work, Team Collaboration